

Embedded Linux & System Software Interview Handbook

Chapter 6 - Device Tree & Platform Drivers (Part 2)

Topics: Device Tree Syntax, Nodes, Properties, Phandles, GPIOs, Interrupts, Clocks

6.5 Device Tree Syntax

A Device Tree is a hierarchical description of the hardware. Each hardware component is represented by a node containing one or more properties. Drivers read these properties during probe() to configure the hardware.

Important Concepts

Concept	Description
Node	Represents a hardware block
Property	Configuration data for a node
compatible	Used for driver matching
reg	Physical address and size
status	okay / disabled
label & phandle	References between nodes

Example UART Node

```
uart0: uart@1000 {  
    compatible = "vendor,uart";  
    reg = <0x1000 0x100>;  
    interrupts = <5>;  
    clocks = <&clk0>;  
    status = "okay";  
};
```

Frequently Used Properties

- compatible - identifies the hardware.
- reg - register base address and size.
- interrupts - IRQ number and trigger information.
- gpios - references GPIO controllers.
- clocks - clock provider.
- resets - reset controller.
- pinctrl-0 - pin multiplexing configuration.

Phandles

Phandles allow one Device Tree node to reference another. For example, a UART node may reference a clock controller or GPIO controller using the '&label;' syntax.

Reserved Memory

The reserved-memory node prevents the kernel from allocating specific RAM regions that are dedicated to firmware, DMA buffers or secure components.

Aliases and Chosen

The aliases node provides stable names such as serial0. The chosen node passes boot arguments, stdout-path and initrd information to the kernel.

Device Tree Compilation

Source files (.dts/.dtsi) are compiled into a binary .dtb using the Device Tree Compiler (dtc), which is loaded by the bootloader together with the kernel.

Common Bring-up Issues

- Wrong compatible string.
- Incorrect reg address.
- Missing interrupt property.
- Disabled node.
- Incorrect GPIO or clock references.
- Bootloader loading the wrong DTB.

Interview Questions

1. What is a phandle?
2. Why is reg important?
3. Explain compatible.
4. What is reserved-memory used for?
5. Difference between aliases and chosen?
6. How do you debug an incorrect Device Tree?

Key Takeaways

Understanding Device Tree syntax is essential for board bring-up and platform driver development. Most driver initialization problems are traced to incorrect Device Tree descriptions.